

Does a firm's takeover vulnerability cause its stock price to deviate from random walks?

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Abstract

This paper examines whether a firm's takeover vulnerability increases the scope of speculative noise trading for its stock. Specifically, the paper tests whether takeover vulnerability *overly* motivates investors to acquire and trade on private information and thus causes them to react even to non-information as if it were information; if so, then the stock price changes will be serially correlated. Using variance ratio, we find that the hypothesized pattern is indeed borne out by data but it is conditional on the quality of information environment. That is, a stock's deviation from random walks is larger when its takeover vulnerability is greater *and* its information environment is poorer. In a sufficiently good information environment, on the other hand, there is no evidence that takeover vulnerability causes speculative noise trading. All those patterns are observed only in individual stock returns and not at the portfolio level. The no-result with portfolios implies that the deviation from random walks observed in individual stock returns is driven by a firm-specific component and thus not explained by time-varying expected returns.

Keywords: Takeover vulnerability; Noise trading; Information environment; Random Walk; Variance ratio

JEL classification: G10; G14; G30; G34

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1. Introduction

Arguably, takeovers are one of the most extreme informational events for stock market investors (e.g., Cao, Chen, and Griffin 2005). It is thus not surprising that a firm's openness to takeover markets is often expected to incentivize investors to acquire and trade on private information and thus contribute to the informativeness of its stock (e.g., Ferreira and Laux 2007). As pointed out by Roll, Schwartz, and Subrahmanyam (2010), however, the conditions for informed trading might also invite "non-informed true believers," since investors engage in trading as long as they believe—right or wrong—that they have private information. That is, takeover vulnerability might also cause some investors to trade on noise as if it were information (e.g., Black 1986; De Long et al. 1990). In this paper, we examine the extent to which a firm's takeover vulnerability increases the scope of such speculative noise trading for its stock.

The sign of takeover-related speculative noise trading is obvious: since noise traders impound non-information into price, the stock price changes will be serially correlated. On one hand, as a non-information-driven price change cannot persist, a negative autocorrelation will arise as the correction occurs. When such correction takes a while, on the other hand, the autocorrelation will first be positive for some short horizons and then become negative for longer intervals. Either way, the stock price implication of takeover-related noise trading is a deviation from random walks. We gauge such a deviation using the standard methodology, namely, the variance ratio (e.g., Lo and MacKinlay 1988), and associate it with a measure of takeover vulnerability based on the number of firm-level anti-takeover provisions.

In doing so, we exercise great caution in the following three dimensions. First, we use high-frequency data, so that the serial correlation of stock returns cannot be attributed to time-variation in expected returns.¹ Second, we focus on a small number of key takeover protections, so that our

¹ In line with this view, Ahn et al. (2002) say that "... because time variation in expected returns is not a high-frequency phenomenon; asset pricing models link expected returns with changing investment opportunities, which, by their nature, are low-frequency events" (p.656). Griffin, Kelly, and Nardari (2010)

takeover vulnerability measure does not represent other correlated firm characteristics, such as the overall governance quality. (We elaborate on this point in the next paragraph.) Third, we explicitly take into account the role of information environment. Griffin, Kelly, and Nardari (2010) show that in using the traditional efficiency measures such as the variance ratio, it is important to control for information environment. In this paper, we do not only control for the difference in information environment across stocks. We also examine whether the hypothesized effect of takeover vulnerability on stock price is conditional on the quality of information environment. An investigation into this conditionality is *ex ante* well motivated. It is because the speculative noise trading hypothesis is premised on the investors' incentive to acquire and trade on private information—be it genuine information or just noise—and this incentive is likely to be stronger in a relatively poorer information environment in which there is some room for “private” information (Grossman and Stiglitz 1980).

Over the period from 1990 to 2005, we calculate four variance ratios for each of our sample stocks. (Our sample is detailed in Section 3.) They are the ratios of 1-day return variance to 5-, 10-, 15-, or 20-day return variance, in which the denominator is scaled by 5, 10, 15, or 20. That is, we examine the serial correlation in daily stock returns over 5, 10, 15, or 20 days at the individual stock level. To measure takeover vulnerability, we focus on three core anti-takeover provisions compiled by the Investor Responsibility Research Center (IRRC) database, namely, classified (staggered) boards, blank check preferred stock, and restrictions on shareholders on calling special meetings or acting through written consent (Cremers and Nair 2005). To create a measure that increases with takeover vulnerability, we deduct from three the number of those anti-takeover provisions within a firm. For each sample stock, this takeover vulnerability measure is averaged over the same period as the variance ratio calculation, and then the average takeover vulnerability

also state that “[G]iven the relatively short (daily or weekly) time horizons we consider, the levels and the differences in autocorrelations are not likely due to time-variation in expected returns” (p.3; ft.4). See also French and Roll (1986).

is associated with the variance ratio cross-sectionally.

We find that both the variance ratio and the absolute value of one minus variance ratio increase, respectively, with our takeover vulnerability measure across all four horizons (i.e., 5, 10, 15, and 20 days). These results mean that the stocks with greater takeover vulnerability deviate more from the random walk process (according to the absolute value of one minus variance ratio result), and further, that the deviation occurs in the form of a negative serial correlation in daily stock returns (according to the variance ratio result). We also find that the relationship between takeover vulnerability and variance ratio is conditional on the quality of information environment. Specifically, the association of takeover vulnerability with a higher variance ratio, and with the ratio's greater deviation from one, is observed only in small stocks or in stocks with a wide bid-ask spread. In a good information environment, on the other hand, such an association disappears completely. Of course, we obtain all those cross-sectional results after controlling for various firm characteristics including the industry effects.

Prior studies employ a broader set of firm-level anti-takeover provisions to measure the overall governance quality. For example, Gompers et al. (2003) construct the so-called governance index using as many as 24 provisions (including the three in our takeover vulnerability measure) to assess the power balance between corporate managers and outside shareholders. Bebchuk, Cohen, and Ferrell (2009) focus only on six of the 24 provisions (including one provision in our takeover vulnerability measure) to more precisely quantify the degree of managerial entrenchment; hence its name, the entrenchment index. Although those broader-based governance measures are correlated with our takeover vulnerability measure, they are not identical. For example, the correlation coefficient between our takeover vulnerability measure and those governance measures is at most 0.609. When one of those governance measures substitutes for our takeover vulnerability measure, the conditionality of the result is noticeably weaker. This weaker result with the governance measures is indirect evidence that takeover vulnerability, not the overall governance quality or

other correlated firm characteristics, is the driver of the conditionality.

Conversely, we employ a direct measure of takeover vulnerability to further sharpen our inferences. Cremers, Nair, and John (2009) estimate a logit regression using actual takeover events and calculate the probability of a firm becoming a takeover target as a function of its various firm characteristics. Although an ex-post measure, this direct measure varies over time for a given firm as the comprising firm characteristics are updated mostly at annual frequencies. This feature enables us to associate *changes* in takeover vulnerability with *changes* in variance ratio, thereby allowing us to establish their causal relationship.² Using panel regressions with firm fixed effects, we find that an increase in takeover vulnerability is related to an increase in variance ratio and, again, that this within-firm result is more pronounced among the stocks in a relatively poorer information environment.

We stress that all the preceding results are found in individual stock returns. We intentionally focus on individual stocks, since the variance ratio of a portfolio would only reveal the average *cross*-autocorrelation within the portfolio. Still, an analysis at the portfolio level will be instructive. More precisely, to the extent that the temporary component detected by the preceding variance ratio analysis is correlated across stocks, the observed pattern—i.e., the conditional relationship between takeover vulnerability and variance ratio—will also be present in the portfolio variance ratios. Or equivalently, the absence of a similar pattern in the portfolio variance ratio would indicate that the temporary component is idiosyncratic. Distinguishing between the two scenarios is crucially important, because the soaring of a takeover target's stock price—the very motivation for the hypothesized speculative trading—is likely to be caused by an increase in future cash flows rather than by a reduction in discount rate. In other words, the speculative noise trading hypothesis posits that the relationship between takeover vulnerability and variance ratio is due to a firm-specific (i.e., cash-flow) component (Vuolteenaho 2002). We confirm this premise, as the variance

² We only use the ratio of 1-day to 5-day return variances in this panel regression. Longer-horizon returns would be too few within a year to provide a reliable variance estimate.

ratios of the portfolios sorted by our takeover vulnerability measure show no particular pattern, regardless of whether the portfolios are equally weighted or value-weighted.

Ferreira and Laux (2007) show that idiosyncratic stock return volatility (as a fraction of the total volatility) is positively related to takeover vulnerability. Noting that private information is likely to be firm-specific rather than market-wide, the authors interpret their result as evidence of greater informed trading in stocks with greater takeover vulnerability. However, the authors also find that the efficiency of corporate capital budgeting decisions—as measured by the marginal q ratio—increases only with the idiosyncratic volatility that is *unrelated* to takeover vulnerability. Given that more information in stock price is likely to help rather than hurt corporate decision making, the marginal q result suggests that takeover vulnerability does not contribute to the stock price informativeness. Therefore, their findings are, at least, not inconsistent with ours. However, it is not easy to directly compare their results with ours, since the former is an unconditional result while the latter is a conditional one.

Indeed, idiosyncratic stock return volatility is one of the most popular measures of stock price informativeness in recent literature (e.g., Morck, Yeung, and Yu 2000; Durnev et al. 2003; Durnev, Morck, and Yeung 2004; Chen, Goldstein, and Jiang 2007). In Section 2.3, we show how the variance ratio is related to the idiosyncratic volatility measure, and also demonstrate that the former is superior to the latter. As a brief summary, while both measures utilize the second moment of stock return, the variance ratio distinguishes information-driven price changes from noise-induced volatility using the simple notion that price changes due to non-information cannot persist. In contrast, the relative idiosyncratic volatility measure is on the grounds that only the idiosyncratic component in stock return volatility is related to private information.

Most prior studies on takeover vulnerability focus on the first moment of stock return and attempt to answer the question of whether takeover vulnerability is, on average, associated with a higher or a lower stock return (e.g., Gompers, Ishii, and Metrick 2003; Core, Guay, and Rusticus

2006; Johnson, Moorman, and Sorescu 2009). This approach is understandable given the axiomatic corporate goal—i.e., maximization of shareholder wealth. While not addressing this issue, our analysis helps us understand another important topic, namely, how a firm’s takeover vulnerability affects the way that new information about the firm arrives in the market and stock market investors react to it. In particular, we show that the openness to takeover markets has the potential to invite noise traders and, perhaps more importantly, that one way of mitigating such an adverse effect of takeover vulnerability is to improve the stock’s information environment.

This paper proceeds as follows. In Section 2, we formally develop our testing hypothesis. Section 3 details our sample and data, and Section 4 reports the empirical results. Section 5 concludes the paper.

2. Hypothesis development

2.1. Setup

To formulate the speculative noise trading hypothesis, we begin by positing that the stock price contains a temporary component. Specifically, we use the following model for stock price (in logarithm):

$$P_t = P_t^* + u_t, \quad (1)$$

where P_t^* follows a random walk process and u_t follows a mean-zero stationary process. The former thus represents a permanent component in stock price (with an innovation, ε_t , following *iid* $(0, \sigma_\varepsilon^2)$), whereas the latter stands for a temporary component (with a variance of σ_u^2). Simple as it is, this model proves to be fairly general. For example, Beveridge and Nelson (1981) show that “any time series which exhibits the kind of homogeneous non-stationarity typical of economic time series can be decomposed into two additive components, a stationary series and a pure random walk” (p.153). The model is also a generalized version of Summers (1986), Fama and French (1988), and Poterba and Summers (1988). As mentioned in the introduction, the nature of a

temporary component can be made unambiguous (empirically) by shortening the measurement interval for price changes.³

In this setup, it is well known that the average one-period return variance is:

$$\text{var}(R_1) = \sigma_e^2 + 2(1 - \rho_1)\sigma_u^2, \quad (2)$$

where R_1 stands for one-period stock return and ρ_1 is the one-period autocorrelation of the temporary component. Similarly, the variance of k -period return is $k\sigma_e^2 + 2(1 - \rho_k)\sigma_u^2$, in which ρ_k is the k^{th} -order autocorrelation of the temporary component (e.g., Poterba and Summers 1988; Campbell, Lo, and MacKinlay 1997). With a sufficiently large k , the variance ratio converges to:

$$\begin{aligned} VR_k &\equiv \frac{\text{var}(R_1)}{\text{var}(R_k)/k} \\ &= \frac{\sigma_e^2 + 2(1 - \rho_1)\sigma_u^2}{\sigma_e^2 + \frac{2}{k}(1 - \rho_k)\sigma_u^2} \\ &\rightarrow \frac{\sigma_e^2 + 2(1 - \rho_1)\sigma_u^2}{\sigma_e^2} \\ &= 1 + \frac{2(1 - \rho_1)\sigma_u^2}{\sigma_e^2}. \end{aligned} \quad (3)$$

A sufficiently large k ensures that a temporary component—i.e., noise—in stock price manifests itself unequivocally as a higher variance ratio. More precisely, Eq. (3) reveals two scenarios in which the temporary component is related to the variance ratio. One is when there are more innovations in the temporary component (i.e., greater σ_u^2), and the other is when a given innovation in the temporary component is more quickly incorporated into price and also corrected fast (i.e., more negative r_1). Put differently, the variance ratio increases—and deviates from one—with the amount of noise and with the speed at which investors respond to noise.

³ In a similar attempt to decompose stock return, Campbell (1991) suggests a vector autoregression (VAR) model with some pre-select state variables. We do not employ this alternative approach, since most candidate proxies for the state variables are observed only at a low frequency. Recall that utilizing high-frequency data is crucial for our purposes, since the nature of a temporary component would otherwise be indeterminate. For the correspondence between Eq. (1) in the text and this alternative VAR specification, see Cohen et al. (2002; p.415-416).

When k is relatively small, on the other hand, the convergence in Eq. (3) does not obtain. Consequently, the ratio can be smaller than one. A closer look at the second line of Eq. (3) reveals the condition for a less-than-one variance ratio, which is $(1 - \rho_1) < (1 - \rho_k)/k$. While three different variables affect this inequality, ρ_1 is likely to play the largest role because ρ_k is typically small in magnitude (even with a relatively small k) and k is given in this discussion. Specifically, ρ_1 is at work in such a way that a more persistent noise—i.e., more positive ρ_1 —makes the stock returns positively autocorrelated and the variance ratio smaller than one. However, the noise in price will dissipate eventually and thus, as k increases (while remaining a small number), the stock returns will eventually be negatively autocorrelated making the ratio greater than one. This non-monotonic pattern is well in line with the U-shaped pattern conjectured by Fama and French (1988; p.251).

2.2. Hypothesis

If the takeover vulnerability of a firm causes speculative noise trading for its stock, then its stock price will deviate from random walks. Specifically, we expect to observe one of the following two patterns:

H1-1: *Greater takeover vulnerability is associated with a higher variance ratio (i.e., a more negative autocorrelation in stock returns).*

H1-2: *Greater takeover vulnerability is first related to a lower variance ratio (i.e., a more positive autocorrelation) for some short horizons, and then it is associated with a higher variance ratio (i.e., a more negative autocorrelation) over longer horizons.*

As mentioned earlier, we hypothesize that takeover vulnerability might cause speculative noise trading because it incentivizes investors to engage in private information acquisition. Quite

naturally, investors will do so only when such an effort rewards them with a comparative informational advantage (Grossman and Stiglitz 1980). Therefore, somewhat imperfect information environments are an important pre-condition for takeover vulnerability to cause speculative noise trading. Consequently, we have the following hypothesis:

H2: *The relationship between takeover vulnerability and variance ratio, as hypothesized by H1-1 and H1-2, are conditional on the quality of information environment. More precisely, their relationship is more pronounced when the information environment is relatively poorer.*

The hypothesized speculative noise trading is aimed at identifying future takeover targets whose stock prices will rise upon the takeover announcement. We note that such a stock price increase is likely to be caused by an increase in future cash flows rather than by a reduction in discount rate. In other words, the hypothesized relationship between takeover vulnerability and variance ratio is caused by a firm-specific component (Vuolteenaho 2002). Thus, we have the following hypothesis:

H3: *The hypothesized relationship between takeover vulnerability and variance ratio is not present at the portfolio level.*

2.3. Relationship between variance ratio and relative idiosyncratic volatility measure

In recent literature, one of the most popular measures of stock price informativeness is the relative idiosyncratic volatility, often known as “one minus R^2 ” (e.g., Morck, Yeung, and Yu 2000; Durnev et al. 2003; Durnev, Morck, and Yeung 2004; Chen, Goldstein, and Jiang 2007; Ferreira and Laux 2007). This relative measure is based on the premise that private information is likely to be firm-specific rather than market-wide; and thus the idiosyncratic volatility but not the systematic

volatility reflects private information flows. However, it is possible—and in fact quite plausible—that private information concerns some market-wide events. For example, the lead-lag relationship between the spot and futures markets for a broad-based stock market index is considered to be evidence of market-level private information held by the futures traders (e.g., Chan 1992).⁴ Moreover, private information acquisition activities for individual firms and for the market as a whole might be positively correlated.⁵ Either way, firm-level private information and market-wide private information would crowd each other out in the relative measure (e.g., Chan and Hameed 2006; Fernandes and Ferreira 2008).

While also based on the second moment of stock return, the variance ratio makes none of those controversial assumptions. To see this more clearly, consider the following two decompositions of stock return variance. First, the return variance of a stock can be viewed as the sum of the variances of a temporary component (denoted below by *temp*) and of a permanent component (denoted by *perm*). Alternatively, the stock return variance can be the sum of the variances of an idiosyncratic component (denoted by *idio*) and of a systematic component in stock return (denoted by *sys*). Thus, we have:

$$\text{var}(\textit{idio}) = w_1 \times \text{var}(\textit{temp}) + w_2 \times \text{var}(\textit{perm}),$$

where w_1 and w_2 are both between 0 and 1, meaning that the systematic component can also contribute to both variances. The relative idiosyncratic volatility, denoted by r_idio , is then:

$$r_idio = \frac{w_1 \times \text{var}(\textit{temp}) + w_2 \times \text{var}(\textit{perm})}{\text{var}(\textit{temp}) + \text{var}(\textit{perm})}.$$

On the other hand, the variance ratio can be rewritten, based on Eq. (3), as follows:

$$VR_k = \frac{\text{var}(\textit{temp}) + \text{var}(\textit{perm})}{\text{var}(\textit{perm})}.$$

⁴ See also Albuquerque, Bauer, and Schneider (2009) for evidence of market-level private information.

⁵ For example, Durnev et al. (2003) say that “... because some business activities are more subject to economy- and industrywide shocks than others, and firm-specific events in these industries may be correspondingly more intense...” (p.801).

Combining the above two equations leads to:

$$r_idio = w_1 \times (1 - VR_k^{-1}) + w_2 \times VR_k^{-1}$$

or

$$VR_k = \frac{(w_2 - w_1)}{(r_idio - w_1)}.$$

With the assumptions embedded in the relative idiosyncratic volatility measure (i.e., $w_1=0$ and $w_2=1$), VR_k is simply the reciprocal of r_idio . However, VR_k does not preclude the possibility that w_1 is positive and w_2 is less than one. The variance ratio draws only on the notion that the impact of non-information on stock price cannot persist and is reversed eventually.

3. Sample and data

Our sample consists of the companies in the IRRC database from which we obtain the information on firm-level anti-takeover provisions.⁶ Following the convention in the literature, we screen out the companies with multiple classes of stock (see, e.g., Gompers et al. 2003; ft.5). Daily stock returns are obtained from the Center for Research in Security Prices (CRSP) and we require at least three years' worth of daily stock return data to be available during our sample period. The sample period starts in September 1990 when the first edition of the IRRC database became available, and it ends in December 2005 as some of the variables in our analysis are no longer available afterwards. Those data requirements leave us with 1,720 firms. For the multivariate analysis, we employ several other databases such as the Compustat. They serve as additional data requirements and reduce the sample size to 1,611 firms.

4. Empirical results

4.1. Summary statistic

⁶ For more detailed information on the IRRC database, we refer the reader to Gompers et al. (2003).

Table 1, Panel A, reports summary statistics of the log return at various measurement intervals, namely, 1 day, 5 days, 10 days, 15 days, and 20 days. Across the 1,720 sample stocks, the average daily log return is virtually zero with a standard deviation of 3.2%. Both the average and the standard deviation increase with the length of the interval. Similarly, daily log returns range widely between -28.5% and 23.1%, and this range increases, albeit somewhat modestly, with the interval length. For example, the 20-day returns are between -44.2% and 37.2%, and it is less than twice the daily return case.

Panel B of Table 1 reports summary statistics of the variance ratio. In calculating the ratio, we only use the k -day windows during which there is no missing daily return. We also use at least 30 observations in estimating the return variance. For example, a valid 20-day return variance needs at least 30 20-day windows each of which has all 20 daily returns. Using these return variances, we calculate four ratios for each stock, namely, VR_5 , VR_{10} , VR_{15} , and VR_{20} .⁷ As shown in the table, both the mean and the median ratios are greater than one for all four k 's. While the minimum variance ratio is as small as 0.52 (for VR_{15}), approximately 65 percent of the sample stocks have a greater-than-one variance ratio. This suggests that our choices of k —i.e., 5, 10, 15, and 20—are long enough to observe the correction of noise, or equivalently, the negative autocorrelation in stock returns.

4.2. Univariate analysis

Table 2, Panel A, reports the mean and median variance ratios for each of the four groups sorted by our takeover vulnerability measure. As mentioned in the introduction, the measure is averaged for each stock across the period during which its variance ratios are calculated.⁸ Group 1

⁷ The estimated variance ratio is biased when the sample is small (e.g., Cochrane 2005). In our case, such a bias is likely to be small, since we use at least three years' worth of data. For example, with 750 daily observations, the bias for VR_5 is of the order of 0.005.

⁸ Information loss associated with the averaging is likely to be small, since the measure does not change much over time. We return to this issue in Section 4.3.5.

contains stocks whose average measure is equal to 3. Group 2 consists of stocks whose average measure is less than 3 but greater than or equal to 1.5. Stocks in group 3 have the average measure less than 1.5 but greater than 0. Finally, group-4 stocks have the average measure of 0. Put simply, group-1 stocks are most vulnerable to takeovers with none of the three core protections during the entire sample period, whereas group 4 has the stocks that are least vulnerable to takeovers with all three protections at all times. Within each of the four groups, the average and median variance ratios are calculated across the constituent stocks.

Two observations stand out in Panel A. First, and consistent with the results in Table 1, all the cross-sectional average and median variance ratios are above one, meaning that the deviation from random walks generally occurs in the form of a negative, not positive, autocorrelation. Second, and more importantly, the variance ratio increases with takeover vulnerability. Across the four values of k , the p -values for the mean difference between groups 1 and 4 are at most 0.032, although the p -values for the median difference are somewhat higher, ranging between 0.021 and 0.136. It is particularly noteworthy that the higher variance ratio of the stocks with greater takeover vulnerability is observed across all the four values of k . This univariate result is thus consistent with H1-1 rather than with H1-2. In other words, the result shows that takeover vulnerability makes investors more vigilant and fast both to react to non-information and to realize their mistake and correct it.

Related, the above-one variance ratios across all four groups facilitate another interpretation. More precisely, the higher variance ratio of stocks with greater takeover vulnerability is automatically read as their greater deviation from random walks. Thus, it is difficult to argue that those higher variance ratios are indicative of a faster correction of—and thus a smaller—mispricing. Similarly, it would be a stretch to claim that the lower variance ratio of less vulnerable stocks is a sign of a more persistent—and thus a larger—mispricing. Such arguments are clearly inconsistent with the larger (smaller) deviation from random walks of the stocks with greater (smaller) takeover

vulnerability.

As emphasized from the beginning of the paper, the speculative noise trading hypothesis predicts that the relationship between takeover vulnerability and variance ratio is conditional on the quality of information environment. Thus, we examine the difference between groups 1 and 4 separately when the information environment is good (Panel B) and when it is bad (Panel C). While there are a number of candidate proxies for the quality of information environment, we use firm size and bid-ask spread. Our idea is that, as readily observable firm characteristics, firm size and bid-ask spread correspond, respectively, to two central building blocks of a stock's information environment—namely, information production and transaction costs (Griffin, Kelly, and Nardari 2010)—reasonably well while not requiring estimation or reducing sample size. Thus, we test the difference between groups 1 and 4 separately with the stocks whose average firm size (or bid-ask spread) is above the sample median and with the stocks whose average firm size (or bid-ask spread) is below the median.⁹

Panels B and C show that the difference in variance ratio between groups 1 and 4 is indeed conditional on the quality of information environment. While the difference between groups 1 and 4 remains significant in a relatively poorer environment (Panel C; small stocks or stocks with a wide bid-ask spread), it is striking that this difference disappears completely when the environment is good (Panel B; large stocks or stocks with a narrow bid-ask spread). The differing results between the sub-samples support the interpretation that the investors' incentive to acquire and trade on private information is the driver of our results.

4.3. Multivariate analysis

Multivariate analysis helps us ensure that the conditional relationship between takeover vulnerability and variance ratio in the preceding univariate analysis is robust to controlling for

⁹ The average firm size is calculated over the same period as that for the variance ratio calculation. The same rule applies to the average bid-ask spread, but it starts only since January 1, 1993 due to data availability.

other firm/stock characteristics. By examining both the variance ratio and the absolute value of one minus variance ratio, we also confirm that the takeover vulnerability-related deviation from random walks occurs in the form of a negative autocorrelation. We first introduce the control variables and then report the regression results.

4.3.1. Control variables

Given the finding of French and Roll (1986) that daily returns on small stocks are negatively autocorrelated, firm size is a natural control variable. A negative autocorrelation in stock return at daily frequency might also be caused by the closing price bouncing between the bid and ask prices (Roll 1984), so the bid-ask spread warrants controlling. We control for book-to-market ratio, since it gauges the extent of far-away cash flows in stock price. Such long-term cash flows add to uncertainty in firm value and thus affect the variance ratio. Similarly, leverage makes equity value more uncertain and thus creates room for noise; and thus we control for it. Since we cannot identify all firm characteristics that affect uncertainty in firm value, we include k -day stock return variance as another control variable. Institutional holdings are included to take into account the investor heterogeneity in terms of their sophistication.

To control for industry effects, we use the 0/1 industry dummy variables based on Fama and French's 49 industry classifications. This control is crucial, since takeover vulnerability and variance ratio are likely to be endogenously related at the industry level. For instance, Gillan, Hartzell, and Starks (2003) show that industry is an important factor affecting a firm's governance including its takeover vulnerability. In particular, Cremers, Nair, and Peyer (2008) show that, conditional on a certain relationship with stakeholders, companies in more competitive industries have more antitakeover provisions. Gaspar and Massa (2006) find that firms in more competitive industries have a greater idiosyncratic volatility. To the extent that such volatility is caused by noise in stock price, the variance ratio in those industries will also be higher.

Table 3, Panel A, reports summary statistics of those control variables, along with those of our takeover vulnerability measure (*ToVul*). Like *ToVul*, each of the control variables is first averaged across the period during which variance ratio is calculated, and the summary statistics in the table are calculated using the cross-section of those averages. The statistics ensure that there are no extreme values in the control variables.

The panel also reports the summary statistics of two proxies for the quality of information environment, denoted by *rk_size* and *rk_Ispread*. Their naming is due to the fact that we transform the original firm size and the bid-ask spread variables into the ranking-based measures. Specifically, we first rank our sample stocks by the average firm size or the inverse of the average bid-ask spread, and then divide the ranking by the number of stocks. The resultant measure ranges between 0 and 1, thereby containing only the information about the cross-sectional rankings. This approach ensures that our results are not sensitive to the specific cross-sectional differences in magnitudes (see, e.g., Llorente et al. (2002; p.1021)). Note also that we use the inverse of the bid-ask spread—and hence denoted by *rk_Ispread*—so that a higher value of this measure represents a better information environment like the firm size-based measure.

At the bottom of Panel A, we report the summary statistics of two broader-based governance measures: *Govern1* that is 24 minus Gompers et al.'s (2003) governance index and *Govern2* that is 6 minus Bebchuk et al.'s (2009) entrenchment index. Panel B shows that those governance measures are highly correlated with our takeover vulnerability measure. At the same time, however, they are not identical to the vulnerability measure, as the correlation coefficient is at most 0.609. The correlation coefficients with the control variables further ensure that our takeover vulnerability measure, *ToVul*, is different from the two governance measures. For example, while *ToVul* is negatively correlated with the transformed bid-ask spread (*rk_Ispread*), both *Govern1* and *Govern2* are respectively positively correlated with it. Also, *ToVul* is not significantly correlated with leverage or book-to-market ratio, whereas *Govern1* and *Govern2* are both significantly correlated

with the two variables. It is thus instructive to substitute those governance measures for our takeover vulnerability measure and see how the results are affected.

The correlation coefficients among the control variables are also significant in most cases and the two proxies for information environment are particularly highly correlated with each other. To facilitate the interpretation of the results, we orthogonalize one proxy to the other (i.e., regressing one proxy on the other and using the regression residuals) to include both variables in one regression.

4.3.2. Main regression analysis

We estimate the following equation:

$$VR_{k,i} = \alpha + \beta_1 ToVul_i + \beta_2 InfoEnv_i + \beta_3 (ToVul_i \times InfoEnv_i) + \sum_{c=1}^C \gamma_c X_{c,i} + \varepsilon_i, \quad (4)$$

where $VR_{k,i}$ is the ratio of 1-day to k -day return variances (the latter being scaled by k) for stock i estimated over the sample period, $ToVul_i$ is the takeover vulnerability of stock i , $InfoEnv_i$ is a proxy for information environment of stock i (either rk_size or rk_Isread), and $X_{c,i}$ represents one of the control variables for stock i . All variables are averaged for each stock across the period during which its variance ratio is calculated. We alternate between the two proxies for information environment (rk_size and rk_Isread). When one variable is used as a proxy, the other variable is first orthogonalized to the former (and denoted by the superscript R) and then is included in the regression as a control variable. Throughout the paper, we use the White (1980) covariance estimator to control for heteroscedasticity.

Table 4 reports the regression results. As shown in the left-half panel, $ToVul$ enters the regression with a significant and positive coefficient, meaning that takeover vulnerability causes a higher variance ratio, or equivalently, a negative autocorrelation in stock returns. More importantly, the interactive term between $ToVul$ and the size-based proxy for information environment

(rk_spread) has a negative coefficient, meaning that the detrimental effect of takeover vulnerability on stock price—i.e., the positive coefficient for *ToVul*—is mitigated by an improvement in information environment. The right-half panel of the table reports the qualitatively identical results. While *ToVul* alone has a positive coefficient, its interactive term with the spread-based proxy for information environment (rk_Ispread) has the negative coefficient. We also note that the results are very similar across the four values of k . Thus, our results again confirm H1-1 but not H1-2.

To sharpen our inferences with a benchmark, we use the absolute value of one minus variance ratio as the dependent variable. More precisely, if takeover vulnerability causes a positive autocorrelation for some stocks and a negative autocorrelation for others, then the results will be weaker with this alternative dependent variable. A similar result to that in Table 4, on the contrary, would confirm that takeover vulnerability causes a deviation from random walks in the form of a negative autocorrelation. Table 5, which reports only the coefficients for *ToVul* and for its interactive term with an information environment proxy, shows that there is no qualitative difference between using the variance ratio and using the absolute value of one minus variance ratio as the dependent variable.

4.3.3. Substituting overall governance measure for takeover vulnerability measure

To ensure that our results are attributable to takeover vulnerability and not to some correlated firm characteristics, we employ a measure that is correlated with takeover vulnerability but is not a measure of takeover vulnerability. Specifically, we use the governance index of Gompers et al. (2003) after deducting it from 24 (*Govern1*) and the entrenchment index of Bebchuk et al. (2010) after deducting it from 6 (*Govern2*). If our results—and the conditionality in particular—are indeed driven by takeover vulnerability, then those governance measures will yield a similar but weaker result.

To save space, we tabulate only the coefficients for a governance measure (*Govern1* or

Govern2) and for its interactive terms with a proxy for information environment (*rk_size* or *rk_Ispread*). Both Table 6 (which uses the variance ratio) and Table 7 (which uses the absolute value of one minus variance ratio) report that the interactive terms of each of the two governance measures with *rk_size* are either insignificant or only marginally significant. This result thus confirms that the conditionality of our earlier results is driven by takeover vulnerability.

4.3.4. Subsample analysis

Thus far, we have focused on the interactive term to detect the conditional relationship between takeover vulnerability and variance ratio. As an alternative, we use our information environment proxies as a discrete variable. Specifically, as in Table 2, we sort the sample stocks into two sub-samples by their average firm size or bid-ask spread and estimate the following equations in each sub-sample:

$$VR_{k,i} = \alpha + \beta ToVul_i + \sum_{c=1}^C \gamma_c X_{c,i} + \varepsilon_i, \quad (5)$$

$$VR_{k,i} = \alpha + \beta Govern1_i + \sum_{c=1}^C \gamma_c X_{c,i} + \varepsilon_i, \quad (6)$$

$$VR_{k,i} = \alpha + \beta Govern2_i + \sum_{c=1}^C \gamma_c X_{c,i} + \varepsilon_i, \quad (7)$$

where *ToVul* is our takeover vulnerability measure, *Govern1* is 24 minus Gompers et al.'s (2003) governance index, *Govern2* is 6 minus Bebchuk et al.'s (2009) entrenchment index, and $X_{c,i}$ is the same set of control variables as the one in Eq. (4).

Again, we save space by reporting only the coefficients for *ToVul*, *Govern1*, and *Govern2*. In both Tables 8 and 9, Panel A reports the results for the good information environment sub-sample and Panel B is for the poor information environment sub-sample. Finally, Panel C reports the *t*-statistics for the difference between Panels A and B. Note first that there is no disagreement between Tables 8 (which uses the variance ratio) and 9 (which uses the absolute value of one minus

variance ratio). Second, and more importantly, the two tables show that the coefficient for *ToVul* is always *insignificant* in the poor information environment sub-samples (Panel A), whereas it is almost always significant in the other sub-samples (Panel B). In stark contrast, the governance measures are even more significantly related to variance ratio—and to the absolute value of one minus variance ratio—when the information environment is good than when it is poor. (Compare the left halves of Panel A and of Panel B). Again, this result confirms that it is takeover vulnerability that drives the conditionality in the results.

4.3.5. Direct measure of takeover vulnerability

As another robustness check, we employ Cremers, Nair, and John's (2009) direct measure of takeover vulnerability, which is the logit estimate of the probability that a firm becomes a takeover target during a given year. More precisely, it is the probability that is based on actual takeover events as a function of the firm's various characteristics measured at the end of the previous year. Specifically, we use the following equation:

$$\begin{aligned} Direct = & -0.083*Q + 0.113*PPE + 0.023*\ln(Cash) - 1.043*BLOCK - 0.072*\ln(Mktcap) + \\ & 0.729*Leverage - 0.527*ROA + 0.048*EXT + 0.053*EXT*BLOCK, \end{aligned}$$

where *Q* is Tobin's *q* ratio, *PPE* is the property, plant, and equipment, *BLOCK* is the institutional holdings with at least 5% ownership, and *EXT* is 24 minus the governance index.¹⁰

As most of the comprising firm characteristics are updated regularly (mostly annually), this direct measure of takeover vulnerability, named *Direct*, varies over time for a given firm. One way of gauging the extent of this time-variation is to compare the within-firm dispersion of this measure with its cross-sectional variation. As reported in Table 10, the average standard deviation of this measure for a given firm is more than half its cross-sectional standard deviation. To put this

¹⁰ Note that, unlike Cremers et al. (2009), we do not include the industry variable to avoid a result stemming from the cross-industry differences. Instead, we control the industry effects using the 0/1 dummy variables, so that the results are driven by the within-industry dispersion.

magnitude into perspective, we repeat the analysis with *ToVul*, *Govern1*, and *Govern2*. The within-firm variation of *ToVul* is only 16% of its cross-sectional variation and this fraction goes up only to 26% with *Govern2*. To take advantage of this within-firm variation of *Direct*, we estimate the following panel regressions with firm fixed effects:

$$VR_{i,t} = \beta_1 Direct_{i,t} + \beta_2 var_{i,t} + FFE_i + \varepsilon_{i,t} \quad (8)$$

$$VR_{i,t} = \beta_1 Direct_{i,t} + \beta_2 var_{i,t} + FFE_i + YFE_t + \varepsilon_{i,t}, \quad (9)$$

where $VR_{i,t}$ is stock i 's variance ratio of 1-day to 5-day stock returns estimated during year t , $Direct_{i,t}$ is stock i 's direct takeover vulnerability measure in year t , $var_{i,t}$ is stock i 's 5-day stock return variance during year t , and finally FFE and YFE are, respectively, firm and year fixed effects. Note that there is no other control variables, since *Direct* itself is a composite variable.

Table 11, first column, shows that *Direct* enters the regression with a significant and positive coefficient in the full sample. This result confirms, again, that greater takeover vulnerability is associated with a higher variance ratio. Given that the firm fixed effects are included in the regression, the result further means that an increase in a firm's takeover vulnerability leads to an increase in the firm's variance ratio. Moreover, the second through fifth columns show that the association between takeover vulnerability and variance ratio is observed only in the sub-samples of small stocks or of the stocks with a wide bid-ask spread. In the other sub-samples, "Large" and "Narrow", *Direct* enters the regression with a significant and negative coefficient. However, Panel B shows that the negative coefficient is not robust to controlling for the year fixed effects, whereas the positive coefficient of *Direct* in "Small" and "Wide" remain significant even after controlling for the year fixed effects.

4.4. Portfolio analysis

We have thus far analyzed individual stock returns. One could ask whether our results carry over to portfolios. To the extent that the "noise" detected by the preceding variance ratio analysis is

correlated across stocks, the observed pattern—i.e., the conditional relationship between takeover vulnerability and variance ratio—will also be present in the portfolio variance ratios. Given that the discount rates, but not the future cash flows, are correlated across individual stocks (Vuolteenaho 2002), such a finding could allow one to argue that the temporary component we find is not a noise but a time-variation in the expected return. To address this issue, we examine the variance ratio of takeover vulnerability-sorted portfolios. To detect the conditional relationship between takeover vulnerability and variance ratio, we construct both equally and value-weighted portfolios. A stronger result with the equally weighted portfolios would be consistent with our earlier individual stock results, since those equally weighted portfolios put more weight on small stocks compared to the value-weighted portfolios.

Table 12, Panels A and B, shows that there is no particular pattern in the variance ratios of the portfolios sorted by takeover vulnerability measure (*ToVul*). Rather, the variance ratios are very similar across all the test portfolios. The no-pattern in the portfolio variance ratios proves that our results with individual stocks are driven by a firm-specific component in stock return. This, in turn, is well in line with the notion that the stock price increase of a takeover target—the very motivation for the hypothesized speculative noise trading—is likely to be caused by an increase in future cash flows rather than by a reduction in discount rate, since the former is idiosyncratic while the latter is not (Vuolteenaho 2002).

Incidentally, we note that all the variance ratios for the portfolios are below one, meaning that their daily returns are positively correlated in time-series. This is in stark contrast to the greater-than-one average variance ratios of individual stock returns (recall the results in Table 2). The less-than-one portfolio variance ratios suggest that stocks in the same portfolio respond to some common shocks, albeit at varying speeds. Such a difference in reaction to a common shock would cause a positive cross-autocorrelation among the stocks and thus make the portfolio variance ratio smaller than one.

5. Conclusions

In this paper, we begin with the notion that a firm's takeover vulnerability creates an incentive for speculative trading of its stock. We then hypothesize that this incentive might cause speculative noise trading, in which case the stock price changes will be serially correlated as non-information is incorporated into price and then corrected. To test this hypothesis, we conduct the variance ratio analysis and find that takeover vulnerability is associated with a deviation from random walks and, further, that this association is conditional on the quality of information environment. That is, a stock's deviation from random walks is larger, the greater is its takeover vulnerability and the poorer is its information environment. In a sufficiently good information environment, on the other hand, there is no evidence that takeover vulnerability causes speculative noise trading. All those patterns are observed only in individual stock returns; at the portfolio level, we find no relationship between takeover vulnerability and the deviation from random walks. Taken together, our results suggest that a stock's takeover vulnerability motivates investors to acquire and trade on private information, but in a poor information environment, takeover vulnerability causes speculative noise trading more than it fosters genuine informed trading.

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Table 1. Summary statistics of k -day log return and variance ratio (VR_k)

Panel A of this table reports summary statistics of k -day log return of the sample stocks ($k = 1, 5, 10, 15$, or 20). The sample consists of the stocks in the Investor Responsibility Research Center (IRRC) database that are not one of the multiple classes of stock for a given firm. Stock returns are obtained from the Center for Research in Security Prices (CRSP) and at least three years' worth of daily stock return data are required during the sample period from September 1990 to December 2005. Panel B of this table reports summary statistics of the ratio of 1-day to k -day return variances (the latter being scaled by k and the ratio being denoted below by VR_k). In calculating the ratio, only the k -day windows during which there is no missing daily return are used. Also at least 30 observations are required in estimating the return variance.

	# firms	mean	stdev	min	median	max
Panel A. k-day log return						
1 day	1,720	0.000	0.032	-0.285	0.000	0.231
5 days	1,720	0.001	0.068	-0.344	0.000	0.298
10 days	1,720	0.001	0.094	-0.389	0.002	0.333
15 days	1,720	0.002	0.115	-0.421	0.003	0.362
20 days	1,720	0.002	0.132	-0.442	0.005	0.372
Panel B. Variance ratio of 1-day to k-day returns (VR_k)						
VR_5	1,720	1.104	0.208	0.556	1.071	3.456
VR_{10}	1,720	1.151	0.286	0.592	1.101	4.562
VR_{15}	1,720	1.175	0.344	0.520	1.107	5.159
VR_{20}	1,720	1.201	0.355	0.560	1.140	4.705

Table 2. Variance ratios by takeover vulnerability-sorted groups

This table reports the mean and median variance ratios of takeover vulnerability-sorted groups. A firm's takeover vulnerability is measured by deducting from three the number of three anti-takeover provisions in the firm, namely, classified (staggered) boards, blank check preferred stock, and restrictions on shareholders on calling special meetings or acting through written consent. This measure is then averaged over the period during which the variance ratio of its stock is calculated. Based on this average takeover vulnerability measure, the sample stocks are assigned to one of the four groups. Specifically, group 1 contains the stocks whose average takeover vulnerability measure is 3 (meaning that none of the three takeover protections exists in the firm during the sample period). Group 2 consists of the stocks whose average takeover vulnerability measure is less than 3 but greater than or equal to 1.5. The stocks in group 3 have the average takeover vulnerability measure that is less than 1.5 but greater than 0. Finally, group-4 stocks have the average takeover vulnerability measure of 0 (meaning that all three takeover protections are present at all times). Panel A reports the results for the full sample, whereas Panels B and C report the results for the sub-samples. To create sub-samples, we use firm size or bid-ask spread that is averaged over the same period as the variance ratio calculation. Due to data availability, the average bid-ask spread is calculated only since January 1, 1993. Panel B is for the stocks whose average firm size (bid-ask spread) is above (below) sample median. Panel C is for the other sub-samples. At the bottom of each panel, we report the *p*-value for the difference in mean or median between groups 1 and 4.

Takeover vulnerability-sorted groups	# firms	mean				median			
		<i>VR</i> ₅	<i>VR</i> ₁₀	<i>VR</i> ₁₅	<i>VR</i> ₂₀	<i>VR</i> ₅	<i>VR</i> ₁₀	<i>VR</i> ₁₅	<i>VR</i> ₂₀
Panel A. Full sample									
1 (most vulnerable)	69	1.17	1.25	1.27	1.32	1.09	1.14	1.16	1.21
2	451	1.11	1.17	1.19	1.23	1.08	1.10	1.11	1.15
3	692	1.10	1.15	1.18	1.21	1.07	1.11	1.11	1.17
4 (least vulnerable)	508	1.09	1.12	1.15	1.15	1.07	1.09	1.09	1.11
<i>p</i> -value for difference: 1 vs. 4		(0.032)	(0.012)	(0.030)	(0.004)	(0.136)	(0.037)	(0.100)	(0.021)
Panel B. Good information environment sub-sample									
1. by firm size (Large)									
1 (most vulnerable)	25	1.08	1.13	1.17	1.26	1.05	1.08	1.11	1.10
4 (least vulnerable)	285	1.08	1.11	1.14	1.14	1.07	1.09	1.11	1.12
<i>p</i> -value for difference: 1 vs. 4		(0.931)	(0.442)	(0.461)	(0.113)	(0.698)	(0.413)	(0.627)	(0.279)
2. by bid-ask spread (Narrow)									
1 (most vulnerable)	19	1.08	1.12	1.18	1.23	1.04	1.08	1.18	1.09
4 (least vulnerable)	293	1.09	1.13	1.16	1.15	1.08	1.11	1.12	1.12
<i>p</i> -value for difference: 1 vs. 4		(0.815)	(0.905)	(0.679)	(0.383)	(0.860)	(0.971)	(0.675)	(0.688)
Panel C. Poor information environment sub-sample									
1. by firm size (Small)									
1 (most vulnerable)	44	1.22	1.31	1.32	1.36	1.11	1.18	1.19	1.23
4 (least vulnerable)	223	1.09	1.14	1.15	1.16	1.06	1.09	1.08	1.09
<i>p</i> -value for difference: 1 vs. 4		(0.024)	(0.023)	(0.046)	(0.021)	(0.047)	(0.067)	(0.091)	(0.057)
2. by bid-ask spread (Wide)									
1 (most vulnerable)	50	1.21	1.29	1.30	1.36	1.09	1.17	1.16	1.23
4 (least vulnerable)	215	1.08	1.11	1.13	1.14	1.05	1.06	1.07	1.09
<i>p</i> -value for difference: 1 vs. 4		(0.020)	(0.006)	(0.024)	(0.006)	(0.062)	(0.008)	(0.048)	(0.012)

Table 3. Summary statistics and correlation coefficients of regression variables

This table reports summary statistics of the regression variables. All variables are first averaged or estimated over the period during which the variance ratio is calculated, and the summary statistics below are calculated using the cross-section of those averages. *ToVul* is our takeover vulnerability measure (3 minus the existence of three anti-takeover provisions in the firm—i.e., classified (staggered) boards, blank check preferred stock, and restrictions on shareholders on calling special meetings or acting through written consent). Two proxies for the quality of a stock's information environment are *rk_size* and *rk_Ispread*, which are, respectively, the transformed firm size (size) and bid-ask spread (spread). Specifically, we first rank our sample stocks by the average firm size or the inverse of the average bid-ask spread. We then divide the ranking by the number of sample stocks. *Govern1* and *Govern2* are, respectively, 24 minus Gompers et al.'s (2003) governance index and 6 minus Bebchuk et al.'s (2009) entrenchment index. A firm's leverage ratio is denoted below by "leverage," its book-to-market ratio by "BM," and the institutional ownership by "Institution."

Panel A. Summary statistics						
Variable	# firms	mean	stdev	min	median	max
<i>ToVul</i>	1,720	1.013	0.871	0.000	1.000	3.000
<i>ln(size)</i>	1,720	20.631	1.501	16.400	20.556	25.957
spread	1,720	0.015	0.016	0.000	0.010	0.178
leverage	1,623	0.225	0.163	0.000	0.218	0.945
<i>ln(BM)</i>	1,613	-0.859	0.677	-6.519	-0.800	2.342
Institution	1,720	0.522	0.254	0.000	0.577	0.965
<i>rk_size</i>	1,720	0.500	0.289	0.001	0.500	0.999
<i>rk_Ispread</i>	1,720	0.500	0.289	0.001	0.500	0.999
<i>Govern1</i>	1,720	15.051	2.534	6.355	15.000	22.000
<i>Govern2</i>	1,720	3.849	1.198	0.000	4.000	6.000

Table 3. cont.

Panel B. Correlation coefficients							
Variable	<i>ToVul</i>	<i>Govern1</i>	<i>Govern2</i>	<i>rk_size</i>	<i>rk_spread</i>	<i>leverage</i>	<i>ln(BM)</i>
<i>Govern1</i>	0.609 (0.000)						
<i>Govern2</i>	0.604 (0.000)	0.714 (0.000)					
<i>rk_size</i>	-0.137 (0.000)	-0.138 (0.000)	-0.022 (0.359)				
<i>rk_spread</i>	-0.150 (0.000)	0.073 (0.003)	0.040 (0.097)	0.562 (0.000)			
<i>leverage</i>	-0.015 (0.548)	-0.109 (0.000)	-0.108 (0.000)	0.002 (0.947)	-0.253 (0.000)		
<i>ln(BM)</i>	0.022 (0.376)	-0.058 (0.020)	-0.074 (0.003)	-0.500 (0.000)	-0.424 (0.000)	0.006 (0.822)	
<i>Institution</i>	-0.056 (0.020)	-0.056 (0.020)	-0.077 (0.002)	0.263 (0.000)	0.360 (0.000)	-0.015 (0.536)	-0.148 (0.000)

Table 4. Regression of variance ratio on takeover vulnerability measure (*ToVul*) and control variables

This table reports the regression results of variance ratio on takeover vulnerability measure (*ToVul*) and control variables. Specifically, Eq. (4) is estimated. As a proxy for the quality of information environment, either the ranking-based firm size (*rk_size*) or the ranking-based inverse bid-ask spread (*rk_Ispread*) is used. (See Table 3 caption for their construction.) When one proxy is used, the other is orthogonalized to the chosen variable as the proxy and then is included in the regression. This orthogonalized variable is denoted by the superscript R. All regression variables are first averaged or estimated over the period during which the variance ratio is calculated, and then included in the regressions. Numbers in brackets are the heteroscedasticity-consistent *t*-statistics. An intercept is in the regression but its coefficient is not reported.

Independent variables	Dependent variable:							
	<i>VR</i> ₅	<i>VR</i> ₁₀	<i>VR</i> ₁₅	<i>VR</i> ₂₀	<i>VR</i> ₅	<i>VR</i> ₁₀	<i>VR</i> ₁₅	<i>VR</i> ₂₀
<i>ToVul</i>	0.053 [3.24]	0.072 [2.68]	0.078 [2.52]	0.095 [2.97]	0.055 [3.10]	0.080 [2.97]	0.090 [2.77]	0.103 [3.51]
<i>ToVul</i> * <i>rk_size</i>	-0.079 [-3.06]	-0.098 [-2.25]	-0.120 [-2.59]	-0.136 [-2.75]				
<i>ToVul</i> * <i>rk_Ispread</i>					-0.082 [-2.64]	-0.114 [-2.53]	-0.144 [-2.79]	-0.151 [-3.13]
<i>rk_size</i>	-0.065 [-1.97]	-0.104 [-2.06]	-0.161 [-2.96]	-0.137 [-2.25]				
<i>rk_Ispread</i> ^R	-0.038 [-1.80]	-0.040 [-1.50]	-0.020 [-0.66]	-0.095 [-3.16]				
<i>rk_Ispread</i>					-0.030 [-0.90]	-0.033 [-0.75]	-0.036 [-0.65]	-0.076 [-1.62]
<i>rk_size</i> ^R					-0.115 [-5.49]	-0.170 [-6.31]	-0.258 [-7.54]	-0.206 [-5.38]
leverage	-0.096 [-2.34]	-0.150 [-3.07]	-0.113 [-1.77]	-0.151 [-2.43]	-0.097 [-2.38]	-0.153 [-3.18]	-0.117 [-1.88]	-0.155 [-2.52]
<i>ln</i> (BM)	-0.021 [-2.68]	-0.037 [-3.74]	-0.067 [-4.10]	-0.062 [-4.70]	-0.022 [-2.79]	-0.039 [-3.85]	-0.069 [-4.24]	-0.064 [-4.78]
Institution	-0.099 [-4.39]	-0.164 [-4.91]	-0.188 [-3.90]	-0.179 [-4.14]	-0.095 [-4.33]	-0.158 [-4.88]	-0.180 [-3.84]	-0.172 [-3.98]
<i>var</i> _k	-5.204 [-3.26]	-5.656 [-5.82]	-5.932 [-10.06]	-4.866 [-9.27]	-5.198 [-3.19]	-5.634 [-5.70]	-5.893 [-9.88]	-4.849 [-9.19]
Industry FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	10.6%	12.3%	13.6%	15.2%	10.7%	12.6%	14.0%	15.5%
# of observations	1,611	1,611	1,611	1,611	1,611	1,611	1,611	1,611

Table 5. Regression of absolute value of (1 – variance ratio) on takeover vulnerability measure (*ToVul*) and control variables

This table reports the regression results of the absolute value of 1 minus variance ratio on takeover vulnerability measure (*ToVul*) and control variables. That is, the regression specification is identical to that for Table 4, except that the dependent variable is the absolute value of one minus variance ratio rather than the variance ratio itself. Numbers in brackets are the heteroscedasticity-consistent *t*-statistics. To save space, we tabulate only the coefficients for takeover vulnerability measure (*ToVul*) and for its interactive term with a proxy for the quality of information environment (*rk_size* or *rk_Ispread*).

Independent variables	Dependent variable: Absolute value of one minus the following VR_k							
	VR_5	VR_{10}	VR_{15}	VR_{20}	VR_5	VR_{10}	VR_{15}	VR_{20}
<i>ToVul</i>	0.046 [3.00]	0.067 [2.68]	0.073 [2.49]	0.085 [2.90]	0.053 [3.51]	0.078 [3.22]	0.087 [2.92]	0.098 [3.63]
<i>ToVul</i> * <i>rk_size</i>	-0.066 [-2.81]	-0.091 [-2.33]	-0.109 [-2.50]	-0.118 [-2.67]				
<i>ToVul</i> * <i>rk_Ispread</i>					-0.080 [-3.25]	-0.114 [-2.92]	-0.135 [-3.01]	-0.144 [-3.37]

Table 6. Regression of variance ratio on overall governance—not takeover vulnerability—measure and control variables

This table reports the regression results of the variance ratio on an overall governance measure and control variables. That is, the regression specification is identical to that for Table 4, except that one of the following two governance measures substitutes for takeover vulnerability measure. One governance measure is 24 minus Gompers et al.'s (2003) governance index (*Govern1*) and the other measure is 6 minus Bebchuk et al.'s (2009) entrenchment index (*Govern2*). Numbers in brackets are the heteroscedasticity-consistent *t*-statistics. To save space, we tabulate only the coefficients for a governance measure (*Govern1* or *Govern2*) and for its interactive term with a proxy for the quality of information environment (*rk_size* or *rk_Ispread*).

Independent variables	Dependent variable:							
	<i>VR₅</i>	<i>VR₁₀</i>	<i>VR₁₅</i>	<i>VR₂₀</i>	<i>VR₅</i>	<i>VR₁₀</i>	<i>VR₁₅</i>	<i>VR₂₀</i>

We tabulate only the coefficients for a governance measure (*Govern1* or *Govern2*) and for its interactive term with a proxy for the quality of information environment (*rk_size* or *rk_Ispread*).

Panel A: Substituting *Govern1* (= 24 – Gompers et al.'s (2003) governance index) for *ToVul*

<i>Govern1</i>	0.014	0.025	0.028	0.030	0.019	0.030	0.034	0.036
	[1.95]	[2.12]	[2.01]	[2.52]	[2.43]	[2.50]	[2.25]	[3.09]
<i>Govern1</i> * <i>rk_size</i>	-0.015	-0.029	-0.033	-0.035				
	[-1.56]	[-1.78]	[-1.81]	[-2.04]				
<i>Govern1</i> * <i>rk_Ispread</i>					-0.026	-0.041	-0.047	-0.049
					[-2.36]	[-2.38]	[-2.21]	[-2.86]

Panel B: Substituting *Govern2* (= 6 – Bebchuk et al.'s (2009) entrenchment index) for *ToVul*

<i>Govern2</i>	0.031	0.047	0.057	0.053	0.039	0.062	0.072	0.070
	[2.27]	[2.36]	[2.39]	[2.42]	[2.98]	[3.34]	[3.06]	[3.41]
<i>Govern2</i> * <i>rk_size</i>	-0.033	-0.052	-0.067	-0.057				
	[-1.78]	[-1.81]	[-2.09]	[-1.82]				
<i>Govern2</i> * <i>rk_Ispread</i>					-0.051	-0.082	-0.098	-0.091
					[-2.86]	[-3.16]	[-2.96]	[-2.94]

Table 7. Regression of absolute value of (1 – variance ratio) on overall governance—not takeover vulnerability—measure and control variables

This table reports the regression results of the absolute value of 1 minus variance ratio on an overall governance measure and control variables. That is, the regression specification is identical to that for Table 5, except that one of the following two governance measures substitutes for takeover vulnerability measure. One governance measure is 24 minus Gompers et al.'s (2003) governance index (*Govern1*) and the other measure is 6 minus Bebchuk et al.'s (2009) entrenchment index (*Govern2*). Numbers in brackets are the heteroscedasticity-consistent *t*-statistics. To save space, we tabulate only the coefficients for a governance measure (*Govern1* or *Govern2*) and for its interactive term with a proxy for the quality of information environment (*rk_size* or *rk_Ispread*).

Independent variables	Dependent variable: Absolute value of one minus the following VR_k							
	VR_5	VR_{10}	VR_{15}	VR_{20}	VR_5	VR_{10}	VR_{15}	VR_{20}

We tabulate only the coefficients for a governance measure (*Govern1* or *Govern2*) and for its interactive term with a proxy for the quality of information environment (*rk_size* or *rk_Ispread*).

Panel A: Substituting *Govern1* (= 24 – Gompers et al.'s (2003) governance index) for *ToVul*

<i>Govern1</i>	0.010	0.020	0.024	0.024	0.015	0.026	0.029	0.029
	[1.86]	[2.08]	[2.03]	[2.34]	[2.38]	[2.65]	[2.21]	[2.98]
<i>Govern1</i> * <i>rk_size</i>	-0.009	-0.021	-0.028	-0.025				
	[-1.24]	[-1.55]	[-1.82]	[-1.73]				
<i>Govern1</i> * <i>rk_Ispread</i>					-0.020	-0.036	-0.040	-0.038
					[-2.15]	[-2.46]	[-2.14]	[-2.69]

Panel B: Substituting *Govern2* (= 6 – Bebchuk et al.'s (2009) entrenchment index) for *ToVul*

<i>Govern2</i>	0.023	0.036	0.048	0.040	0.036	0.053	0.061	0.060
	[2.26]	[2.26]	[2.42]	[2.10]	[3.52]	[4.00]	[3.12]	[3.62]
<i>Govern2</i> * <i>rk_size</i>	-0.024	-0.039	-0.056	-0.036				
	[-1.57]	[-1.57]	[-2.03]	[-1.30]				
<i>Govern2</i> * <i>rk_Ispread</i>					-0.049	-0.073	-0.083	-0.078
					[-3.23]	[-3.72]	[-2.92]	[-2.98]

Table 8. Regressions by sub-samples

This table reports the regression results of variance ratio on takeover vulnerability measure (*ToVul*) or governance measure (*Govern1* or *Govern2*) and control variables. Specifically, Eqs. (5), (6), and (7) are estimated within each of the sub-samples sorted by the average firm size or average bid-ask spread. Panel A is for the stocks whose average firm size (bid-ask spread) is above (below) sample median, and Panel B is for the other sub-samples. To save space, we tabulate only the coefficients for *ToVul*, *Govern1*, and *Govern2*. Numbers in brackets are the heteroscedasticity-consistent *t*-statistics. Finally, Panel C reports the *t*-statistics for the difference in the coefficients for *ToVul*, *Govern1*, and *Govern2* between the sub-samples.

Independent variables	Dependent variable:							
	VR_5	VR_{10}	VR_{15}	VR_{20}	VR_5	VR_{10}	VR_{15}	VR_{20}
	Sub-samples sorted by firm size				Sub-samples sorted by bid-ask spread			
Panel A. Good information environment sub-sample								
$ToVul$	0.002 [0.27]	0.011 [1.35]	0.005 [0.64]	0.014 [1.35]	0.000 [-0.01]	0.005 [0.57]	-0.004 [-0.41]	0.003 [0.25]
$Govern1$	0.005 [3.31]	0.008 [4.32]	0.009 [3.46]	0.011 [3.63]	0.003 [1.94]	0.005 [2.12]	0.005 [1.53]	0.004 [1.05]
$Govern2$	0.009 [2.22]	0.013 [2.52]	0.015 [2.14]	0.018 [2.65]	0.004 [1.13]	0.007 [1.24]	0.007 [1.04]	0.007 [0.90]
Panel B. Poor information environment sub-sample								
$ToVul$	0.024 [2.35]	0.032 [1.98]	0.029 [1.52]	0.038 [1.98]	0.027 [2.80]	0.039 [2.92]	0.039 [2.37]	0.051 [3.05]
$Govern1$	0.009 [1.67]	0.015 [1.75]	0.016 [1.54]	0.015 [1.86]	0.008 [1.61]	0.013 [1.92]	0.014 [1.67]	0.017 [2.58]
$Govern2$	0.018 [1.94]	0.025 [1.84]	0.031 [1.79]	0.026 [1.79]	0.020 [2.28]	0.030 [2.60]	0.034 [2.32]	0.035 [2.76]
Panel C. Difference between Good (Panel A) and Poor (Panel B) information environment sub-samples								
$ToVul$	[-1.86]	[-1.15]	[-1.15]	[-1.11]	[-2.38]	[-2.19]	[-2.27]	[-2.31]
$Govern1$	[-0.57]	[-0.79]	[-0.68]	[-0.49]	[-0.94]	[-1.12]	[-1.07]	[-1.77]
$Govern2$	[-0.90]	[-0.80]	[-0.86]	[-0.49]	[-1.65]	[-1.78]	[-1.62]	[-1.87]

Table 9. Regressions by sub-samples – using absolute value of (1 – variance ratio) as dependent variable

This table reports the regression results of the absolute value of 1 minus variance ratio on takeover vulnerability measure (*ToVul*) or governance measure (*Govern1* or *Govern2*) and control variables. That is, the regression specification is identical to that for Table 8, except that the dependent variable is the absolute value of one minus variance ratio rather than the variance ratio itself. Panel A is for the stocks whose average firm size (bid-ask spread) is above (below) sample median, and Panel B is for the other sub-samples. To save space, we tabulate only the coefficients for *ToVul*, *Govern1*, and *Govern2*. Numbers in brackets are the heteroscedasticity-consistent *t*-statistics. Finally, Panel C reports the *t*-statistics for the difference in the coefficients for *ToVul*, *Govern1*, and *Govern2* between the sub-samples.

Independent variables	Dependent variable: Absolute value of one minus							
	VR_5	VR_{10}	VR_{15}	VR_{20}	VR_5	VR_{10}	VR_{15}	VR_{20}
	Sub-samples sorted by firm size				Sub-samples sorted by bid-ask spread			
Panel A. Good information environment sub-sample								
<i>ToVul</i>	0.004 [0.73]	0.009 [1.31]	0.004 [0.54]	0.010 [1.10]	-0.002 [-0.30]	0.002 [0.38]	-0.002 [-0.33]	0.004 [0.44]
<i>Govern1</i>	0.005 [3.62]	0.006 [3.49]	0.005 [2.20]	0.009 [3.36]	0.002 [1.51]	0.004 [1.94]	0.004 [1.74]	0.004 [1.42]
<i>Govern2</i>	0.009 [3.45]	0.009 [2.48]	0.010 [1.93]	0.017 [3.90]	-0.001 [-0.15]	0.001 [0.29]	0.004 [0.79]	0.005 [0.97]
Panel B. Poor information environment sub-sample								
<i>ToVul</i>	0.020 [2.06]	0.030 [1.97]	0.031 [1.67]	0.037 [2.08]	0.026 [3.17]	0.037 [3.11]	0.038 [2.45]	0.045 [3.07]
<i>Govern1</i>	0.007 [1.66]	0.013 [1.96]	0.015 [1.73]	0.014 [1.99]	0.007 [1.80]	0.012 [2.26]	0.012 [1.70]	0.014 [2.71]
<i>Govern2</i>	0.012 [1.65]	0.019 [1.89]	0.026 [1.88]	0.021 [1.80]	0.019 [3.03]	0.025 [3.45]	0.028 [2.49]	0.030 [3.37]
Panel C. Difference between Good (Panel A) and Poor (Panel B) information environment sub-samples								
<i>ToVul</i>	[-1.48]	[-1.29]	[-1.38]	[-1.38]	[-2.79]	[-2.59]	[-2.37]	[-2.36]
<i>Govern1</i>	[-0.37]	[-1.02]	[-1.10]	[-0.68]	[-1.20]	[-1.48]	[-1.12]	[-1.84]
<i>Govern2</i>	[-0.35]	[-0.97]	[-1.07]	[-0.30]	[-2.72]	[-2.94]	[-1.92]	[-2.41]

Table 10. Summary statistics of a direct measure of takeover vulnerability (*Direct*)

This table reports summary statistics of a direct measure of takeover vulnerability (*Direct*). The measure is Cremers, Nair, and John's (2009) logit estimate of the probability that a firm becomes a takeover target during a given year as a function of the firm's various characteristics measured at the end of the previous year. We first calculate this measure each year for each firm and then average it across the period during which the variance ratio is calculated for its stock. The summary statistics in Panel A and the cross-sectional standard deviation in Panel B are calculated using the cross-section of those averages. The within-firm standard deviation in Panel B is the cross-sectional average of the standard deviation for a given firm.

Panel A. Summary statistics of *Direct*

variable	# firms	mean	stdev	min	median	max
<i>Direct</i>	1,599	0.043	0.322	1.208	0.042	-1.310

Panel B. Comparison between *Direct*, *ToVul*, *Govern1*, and *Govern2*

variable	within-firm stdev (A)	cross-sectional stdev (B)	A / B
<i>Direct</i>	0.172	0.322	53.5%
<i>ToVul</i>	0.135	0.871	15.5%
<i>Govern1</i>	0.605	2.534	23.9%
<i>Govern2</i>	0.310	1.198	25.8%

Table 11. Firm fixed-effects panel regressions of variance ratio on *Direct* and control variables

This table reports the panel regression results of the variance ratio of 1-day to 5-day returns, estimated each year, on the direct measure of takeover vulnerability (*Direct*) and 5-day variance, both calculated during the same year. Firm (and year) fixed effects are also included in the regressions. Specifically, we estimate Eq. (8) for Panel A and Eq. (9) for Panel B. Numbers in brackets are the heteroscedasticity-consistent *t*-statistics. Sub-samples are sorted by the average firm size using the sample median (“Large” and “Small”) or by the average bid-ask spread using the sample median (“Narrow” and “Wide”). At the bottom of each panel, we report the *t*-statistics for the difference in the coefficient for *Direct* between the sub-samples.

Dependent variable: VR_5 estimated each year					
	estimated for full sample	estimated within sub-sample of:			
		Large	Small	Narrow	Wide
Panel A. Regressions with firm fixed effects					
Direct	0.041	-0.034	0.155	-0.052	0.156
	[2.46]	[-2.20]	[4.65]	[-3.19]	[5.11]
var_5	-12.232	-11.533	-12.955	-10.285	-14.524
	[-11.81]	[-7.57]	[-9.24]	[-12.58]	[-7.07]
Firm FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	No	No	No	No	No
R-squared	34.5%	20.3%	41.3%	22.1%	39.6%
# of observations	13,048	7,377	5,671	6,602	6,446
<i>t</i> -statistic for difference in <i>Direct</i> coefficient between sub-samples		[-5.14]		[-6.01]	
Panel B. Regressions with firm fixed effects and year fixed effects					
Direct	0.062	0.004	0.143	-0.019	0.161
	[3.71]	[0.22]	[4.44]	[-1.11]	[5.21]
var_5	-15.396	-16.266	-14.958	-15.063	-16.058
	[-10.41]	[-6.61]	[-7.97]	[-10.71]	[-6.23]
Firm FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes
R-squared	37.4%	25.8%	43.6%	28.1%	41.7%
# of observations	13,048	7,377	5,671	6,602	6,446
<i>t</i> -statistic for difference in <i>Direct</i> coefficient between sub-samples		[-3.82]		[-5.10]	

Table 12. Variance ratios of equally or value-weighted portfolios sorted by takeover vulnerability (*ToVul*)

This table reports the variance ratios of the equally or the value-weighted portfolios sorted by takeover vulnerability. (See Table 2 caption for the construction of our takeover vulnerability measure, namely, *ToVul*.) Each day, stocks are sorted into one of the four portfolios based on their most recent takeover vulnerability measure. Specifically, portfolio 1 contains the stocks whose takeover vulnerability measure is 3 (meaning that they are most vulnerable to takeovers). Portfolio 2 consists of the stocks whose takeover vulnerability measure is less than 3 but greater than or equal to 1.5. The stocks in portfolio 3 have the takeover vulnerability measure that is less than 1.5 but greater than 0. Finally, portfolio-4 stocks have the takeover vulnerability measure of 0 (meaning that they are least vulnerable to takeovers). Portfolio returns are first calculated using simple returns of the constituent stocks, and the variance ratios of the portfolios are calculated using the natural log transformation of the simple portfolio returns—i.e., natural log of (1 + simple return).

Takeover vulnerability-sorted portfolios	VR_5	VR_{10}	VR_{15}	VR_{20}
Panel A. Equally weighted portfolios				
Pf 1 (most vulnerable)	0.90	0.81	0.79	0.79
Pf 2	0.90	0.82	0.79	0.78
Pf 3	0.91	0.80	0.78	0.79
Pf 4 (least vulnerable)	0.90	0.79	0.79	0.78
Panel B. Value-weighted portfolios				
Pf 1 (most vulnerable)	0.91	0.82	0.80	0.81
Pf 2	0.91	0.84	0.81	0.81
Pf 3	0.93	0.82	0.80	0.81
Pf 4 (least vulnerable)	0.91	0.81	0.81	0.80